

Gridworks EMBR

Cymcap 8.1 Validation Report

Validation of EMBR's ampacity engines against Cymcap 8.1 Rev 2

Tool: Gridworks EMBR (Engineered Model for Buried-cable Ratings), v1.0

Reference: Cymcap 8.1 Rev 2 (IEC 60287 / finite-element cable rating software)

Methodology: IEC 60287-1-1 and IEC 60287-2-1 / Neher–McGrath

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1 Purpose and summary

This report documents the validation of the Gridworks EMBR cable ampacity engines against Cymcap 8.1 Rev 2, a widely used commercial cable-rating package. EMBR was run across a validation matrix spanning its three engines and compared, scenario by scenario, against Cymcap target ampacities.

EMBR agrees with Cymcap within $\pm 5\%$ on every one of the 58 validated scenarios (58 / 58).

2 Reference

Cymcap is an established commercial tool for underground cable current-rating calculations, implementing the IEC 60287 / Neher–McGrath methodology with finite-element external-thermal solutions. Version 8.1 Rev 2 was used as the comparison reference. Because Cymcap is independently developed and widely adopted in the transmission and distribution industry, agreement against it provides strong evidence that EMBR's engines are implemented correctly across a realistic range of installations.

3 Scope and methodology

The validation matrix exercises EMBR's three engines across a representative range of cable sizes, installation types (direct-buried and in-conduit), soil thermal resistivities and temperatures, load factors, and single- and multi-circuit arrangements. For each scenario, identical inputs were supplied to EMBR and to a Cymcap model, and the resulting ampacities were compared.

Acceptance criterion: EMBR ampacity within $\pm 5\%$ of the Cymcap target for the scenario.

Unit convention: Soil thermal resistivity in the matrix is expressed in $^{\circ}\text{C}\cdot\text{cm}/\text{W}$; EMBR works internally in $\text{K}\cdot\text{m}/\text{W}$ (divide by 100). The frontend performs this conversion automatically.

4 Results by engine

Every validated scenario passes within $\pm 5\%$.

Engine	Scenarios	Pass ($\pm 5\%$)
MV Trefoil (35 kV)	28	28 / 28 (100%)
DC (solar / battery)	19	19 / 19 (100%)
LVAC (feeders)	11	11 / 11 (100%)
Total	58	58 / 58 (100%)

Result: PASS — 58 / 58 validated scenarios within $\pm 5\%$.

5 Notes

Per-scenario detail: this report summarises the validation by engine. The full scenario-by-scenario matrix (individual cable parameters, Cymcap target ampacities, EMBR results, and delta percentages) is maintained separately in the EMBR validation matrix workbook.

6 Conclusion

Across a validation matrix spanning its MV, DC, and LVAC engines, EMBR reproduces Cymcap 8.1 Rev 2 ampacities within $\pm 5\%$ on all 58 validated scenarios. On this basis EMBR's ampacity engines are validated as consistent with Cymcap across the range of installations covered by the matrix.

7 References

- [1] Cymcap 8.1 Rev 2 — underground cable current-rating software.
- [2] IEC 60287-1-1 and IEC 60287-2-1, “Electric cables — Calculation of the current rating.”
- [3] J. H. Neher and M. H. McGrath, “The calculation of the temperature rise and load capability of cable systems,” AIEE, 1957.
- [4] Gridworks EMBR Cymcap validation matrix and “Cymcap Model Corrections” (internal).